

Rope Cosmology

Cosmology of an Interpenetrating Rope Medium: From the Microscopic Network to the Observable Universe

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Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

This paper develops the cosmological sector of the Mesh Programme. Prior papers describe local physics — electromagnetism, gravity, particle masses, chemistry, thermodynamics, and galactic dynamics — but none addresses the large-scale evolution of the rope medium itself. We establish a cosmological framework: the state variables of a cosmological rope network, candidate microscopic meanings of expansion, conservation laws, and a configuration-entropy account of cosmic evolution. We are explicit from the outset that this is a FRAMEWORK paper, not a results paper: for the classic cosmological problems — dark energy, inflation, the CMB, nucleosynthesis — the honest current status of the rope medium is a set of candidate mechanisms, none yet derived, and we label them as such rather than claiming explanations. The paper's substantive contribution is threefold: (i) it poses the cosmological questions in well-defined rope-medium terms; (ii) it identifies what observations would falsify the picture; and (iii) it shows that if cosmic expansion corresponds to relaxation of rope tension, the programme's existing prediction of correlated α -G variation becomes a COSMOLOGICAL prediction — a genuine, and genuinely risky, testable consequence. Quantitative comparison with cosmological data is reserved for future work.

What this paper does — and does not — do

DOES: define the cosmological rope state; enumerate candidate mechanisms for expansion, dark energy, and inflation with their distinguishing signatures; establish conservation-law structure; extend the configuration-entropy framework to cosmology; state falsification conditions; and connect cosmic expansion to the existing α -G prediction.

DOES NOT: derive the expansion history, explain dark energy or inflation, compute a CMB power spectrum, or predict primordial abundances. Where a classic result is discussed, the rope-medium content is candidate mechanisms only, explicitly labelled. No quantitative fit to cosmological data is claimed.

1. Introduction: The Remaining Foundational Layer

The Mesh Programme has developed foundations for ontology, particle physics, gravity, electromagnetism, and — most recently — the thermodynamics and statistical mechanics of the rope medium. One foundational layer remains: cosmology, the large-scale evolution of the rope network itself. A theory that proposes the substrate of spacetime is, in a sense, obligated to say something about how that substrate evolves on the largest scales — or at least to pose the question precisely and say honestly what it can and cannot yet answer.

We take the second, more modest path deliberately. Cosmology is where speculative substrate theories most often overreach, attaching a qualitative story to every observation without derivation. The discipline that has served the rest of the programme — separating what is derived from the ontology, what is inherited from effective field theory, what is assumed, and what is open — is most important precisely here, where the temptation to overclaim is greatest. This paper is therefore written as a scoping and framework document. Its value, if it has any, is in posing the right questions in rope-medium terms and identifying what would count as answers or refutations.

2. The Cosmological Principle for a Rope Medium

Standard cosmology assumes the universe is, on large scales, statistically homogeneous and isotropic in its matter content. The rope-medium analogue replaces homogeneous MATTER with statistically homogeneous rope PROPERTIES: on scales large compared with the correlation length, the average rope orientation, rope density, rope tension, and defect density are position-independent in a statistical sense.

This is a genuine reformulation, not a cosmetic one. Isotropy of matter becomes isotropy of the orientation distribution (no preferred rope direction after averaging); homogeneity becomes spatial constancy of the statistical parameters. It is also immediately falsifiable in principle: a detected large-scale preferred direction in the rope orientation field would violate it, and such anisotropy could in principle show up in the polarisation observables discussed in §9 and §14.

3. The Cosmological Rope State

We take the cosmological state of the rope medium to be specified, at the coarse-grained level, by a small set of statistical fields:

- $\rho_R(t)$ — the rope density (ropes per unit volume);
- $T(t)$ — the average strand tension (the same primitive that sets the EM coefficient and the gravitational coupling in earlier papers);
- $n(x,t)$ — the average orientation field (a director), whose statistics encode isotropy;
- $D(t)$ — the defect density (density of vortex/monopole topological defects);
- $\xi(t)$ — the correlation length, the scale beyond which orientations decorrelate.

These are the cosmological reductions of quantities defined microscopically in the ontology and thermodynamics papers. The central claim of the framework is that the observable large-scale history of the universe is the joint evolution of these few fields — and, crucially, that they are not independent: T appears in both the electromagnetic and gravitational couplings, so its cosmic evolution links sectors (§13, §14).

Collecting these, the cosmological state of the rope medium is a single vector,

$$R(t) = (\rho_R, \quad T, \quad n, \quad D, \quad \xi) \tag{3.1}$$

and cosmology, in this framework, is nothing more or less than the evolution of that vector under some dynamics F to be determined:

$$dR/dt = F(R) \tag{3.2}$$

Equation (3.2) is the central dynamical object of the cosmological sector. This paper does NOT derive F — that is the task of the successor paper (§17) — but naming the object precisely is what converts “rope cosmology” from a collection of topics into a single well-posed problem: find the evolution law for R . Every candidate mechanism for expansion (§4), and every conservation law (§5), is a statement about the structure of F .

4. What Is Cosmic Expansion, Microscopically?

In standard cosmology space itself expands. In a rope-medium picture there is no space apart from the network, so “expansion” must be some evolving property of the network. We enumerate the candidate mechanisms WITHOUT assuming any one is correct — this is a menu of hypotheses with distinguishing signatures, not a claim:

Candidate mechanism	What expands	Distinguishing signature
Increasing rope spacing	the lattice constant a grows	dilutes density; changes EM coefficient $c = \kappa a/(2T^2)$ over time
Relaxing rope tension	T decreases with time	directly drives α – G variation (§14); links expansion to constant-drift
Decreasing defect density	D falls as defects annihilate	an early defect-dominated era; entropy/structure consequences
Network reorganisation	topology rearranges at fixed density	expansion without dilution; subtler observational imprint

Why this matters for the rest of the programme. The tension-relaxation candidate is not neutral: because $\alpha \sim 2T^2/(\kappa a)$ and the gravitational coupling also depends on T , an expansion that relaxes T would make the fundamental constants evolve over cosmic time in a correlated way. This turns the choice of mechanism into an observational question (§14), which is exactly the kind of leverage a framework paper should expose rather than hide.

5. Conservation Laws and Cosmological Equations

Even absent a derived expansion history, the rope medium must obey conservation laws, and these constrain the evolution more than a purely qualitative story would. We posit rope analogues of the standard cosmological continuity relations, as a structural framework to be filled in:

- Energy: the total network energy (tension + bond + orientation + defect, as in the thermodynamics paper) evolves by the work done as the network reorganises.
- Topology: strand-linking number is conserved locally (charge conservation, from the electricity paper); its cosmological version constrains how the net topological content of the universe can change.
- Defect number: defects are created/annihilated in pairs, giving a defect continuity equation $dD/dt = (\text{creation}) - (\text{annihilation})$, central to the inflation and structure discussions.

These are written as a framework, not solved. The point is that the cosmological sector is not unconstrained storytelling: conservation of energy, topology, and defect number restricts the admissible histories, and any candidate expansion mechanism (§4) must respect them. A mechanism that violated topological conservation, for instance, would be ruled out on internal grounds before any observation.

6. Entropy of the Universe as Configuration Degeneracy

The thermodynamics paper defined the entropy of a macroscopic field as $S = k \ln|C(A)|$, the logarithm of the number of microscopic rope configurations rendering it. Applied cosmologically, the entropy of the universe is the degeneracy of microscopic rope networks compatible with the observed large-scale state. This gives a concrete, if not yet computable, definition of cosmic entropy in rope terms and a natural arrow of time: evolution proceeds toward larger-degeneracy network configurations, subject to the conservation laws of §5.

This framing has a useful consequence. The observed low entropy of the early universe becomes, in rope language, a statement that the initial network occupied a SMALL-degeneracy configuration — a highly ordered or highly constrained network. Section 10 takes up what such an initial state might be. We claim no derivation of the early low-entropy condition; we note only that the mesh framework restates the puzzle in configuration-counting terms rather than dissolving it.

7. Dark Energy: Candidate Mechanisms

Dark energy — the observed accelerating expansion — is the sharpest test of any cosmological framework, and the place a substrate theory is most tempted to overclaim. We therefore list candidate rope-medium sources of a dark-energy-like effect with their signatures, and derive none. Each is a hypothesis to be developed or falsified, not a result.

Candidate source	Rope-medium picture	Would be distinguished by
Residual network tension	an irreducible background tension acting	a specific, near-constant equation-of-

	as a negative-pressure term	state $w \approx -1$
Defect pressure	the collective pressure of residual topological defects	w evolving with defect density $D(t)$; a $w \neq -1$ today
Topological vacuum energy	energy of the network's ground-state topology	a cosmological-constant-like term; the usual magnitude problem persists
Configuration entropy / entropic force	an entropic contribution driving the network toward higher degeneracy	a thermodynamically-derived w , potentially time-dependent

Honest status. None of these is derived. Two of them (defect pressure, configuration entropy) predict a TIME-DEPENDENT equation of state $w(t) \neq -1$, which is observationally distinguishable from a pure cosmological constant — so the framework at least identifies where a rope explanation would differ from Λ CDM. But the notorious magnitude problem (why dark energy has the tiny value it does) is not addressed by any of them, and we do not pretend otherwise.

8. Inflation: Could the Network Produce an Early Rapid Epoch?

We assess, without asserting, whether a rope network could produce an inflation-like early epoch of rapid expansion. Three candidate mechanisms suggest themselves: rapid relaxation of an initially over-tensioned network; a burst of defect annihilation releasing energy; or topological unwinding of an initially highly-knotted network. Each would inject energy and drive rapid reorganisation.

The honest status is that none of these has been shown to give the exponential expansion, the e-fold count, or the near-scale-invariant perturbation spectrum that inflation requires and that observations of the CMB strongly constrain. What the rope picture offers here is a menu of possible drivers and a criterion — any viable rope-inflation mechanism must, after the fact, reproduce the measured spectral tilt and amplitude of primordial fluctuations. We record this as an open problem with a sharp acceptance test, not as a solution.

9. The Cosmic Microwave Background

In standard cosmology the CMB is the relic radiation of last scattering, when the universe became transparent. A rope interpretation must identify what “last scattering” corresponds to in the network — plausibly a transition in the defect density or correlation length past which electromagnetic phase-waves propagate freely. We develop this only far enough to identify possible observable signatures, and explicitly do not claim a complete account or a computed power spectrum.

The most promising rope-specific signature is polarisation. If the cosmological rope orientation field carries any residual large-scale structure or handedness, it could imprint a specific birefringence or polarisation-rotation pattern on the CMB — connecting to the cosmological birefringence prediction already in the falsifiable-predictions paper. This is the natural place for the cosmology sector to make contact with data, and we flag it as the priority target for future quantitative work rather than a present result.

10. The Initial State: What the Network Was, Not What Exploded

The mesh framework reframes the initial-conditions question. Instead of asking what exploded, it asks what the initial rope network looked like. This is a more tractable question because it is about a configuration, not an event. We consider candidate initial networks:

- Tangled: a maximally-knotted network that has been unwinding ever since (pairs naturally with the topological-unwinding inflation candidate).
- Critical: a network poised at the phase boundary (§9 of the thermodynamics paper), which would give scale-invariant fluctuations — attractive, since scale-invariance is observed.
- Maximum-entropy: the network began in its highest-degeneracy state (in tension with the observed low early entropy — §6).
- Minimum-entropy: a highly ordered initial network (consistent with the low early entropy, but requires explaining why).
- Nucleated: the network came into being by a nucleation event from no-network.

The critical-network candidate is the most interesting because it connects an initial condition to an OBSERVABLE (scale-invariant primordial fluctuations) through the known physics of critical phenomena. We highlight it as the most promising line, while claiming no derivation.

10A. A Cyclic Rope Cosmology

The initial-state options above treat the beginning as a single event. A natural alternative is that there is no unique beginning: the rope network OSCILLATES, each cycle's starting configuration being the endpoint of the previous cycle's collapse. This subsection develops that possibility, because it resolves a difficulty the single-shot pictures leave open — the origin of the initial tension — and because it makes the assumed high-tension initial state a CONSEQUENCE rather than a postulate.

The cycle, in rope-medium terms. During expansion the tension T relaxes and the network spreads (§4's tension-relaxation reading). If tension-mediated gravity eventually overcomes the relaxation, expansion turns around and the network re-compresses. At maximal compression the entire network is driven to extreme tension and defect density — which, in the programme, is precisely a black-hole configuration (an extreme-tension state). The compression stores enormous elastic and topological energy; past a limiting compression of the interpenetrating network, that stored energy acts as a restoring force and the collapse reverses — a bounce — launching the next expansion from a high-tension, high-defect, near-critical state. That state is exactly the “initial network” drawn in the phase diagram (§13B), now GENERATED by collapse rather than assumed.

Why this is attractive. Three problems of the single-shot pictures ease at once. First, the origin of tension: it is regenerated at each bounce by compression, so no external tension source is needed.

Second, the high-tension initial state stops being an arbitrary assumption and becomes the derived output of collapse. Third, the tension history becomes naturally NON-MONOTONIC — regenerated at the bounce, relaxing through expansion — which is the third option raised when we asked whether the universe began taut or slack: it began taut in OUR cycle because a previous cycle’s collapse made it so.

The bounce mechanism. “So much compression that it explodes” is made precise as a topological-elastic limit: interpenetrating ropes at maximal tension and maximal knotting cannot be compressed indefinitely; the stored tension and topological energy provide a restoring force. This is a genuine candidate bounce mechanism rather than a singularity — the network need never reach infinite density, only maximal-but-finite compression. We label it a candidate, not a derived result.

The make-or-break problem: entropy. Standard oscillating universes fail on the second law: configuration entropy $S = k \ln|C|$ accumulates each cycle, so the cycles cannot have run for infinite past time (they would lengthen and degrade without bound). The mesh framework has a distinctive — but unproven — potential escape. If the collapse to a single maximal-tension black-hole configuration is a LOW-degeneracy state (small $|C|$), then the bounce could RESET the configuration entropy rather than accumulate it. Whether compression genuinely lowers $|C|$ is a concrete, in-principle-calculable question, and it is the hinge on which the whole cyclic picture turns:

- If collapse RESETS entropy ($|C|$ driven small at maximal compression): the model can be genuinely past-infinite — true eternal oscillation, with no first cycle and no initial-condition problem at all.
- If entropy ACCUMULATES ($|C|$ not reset): the cycling had a beginning; there was a first cycle, and the initial-condition question returns for that first cycle (though later cycles are still explained).

We do not claim to know which holds. Establishing the sign of the entropy change through collapse is recorded as the decisive open problem for cyclic rope cosmology — and, encouragingly, it is the kind of configuration-counting question the programme’s thermodynamic layer is built to pose precisely.

Relation to the α -G prediction and to observation. Because tension is non-monotonic across a cycle, the correlated-drift prediction $d \ln \alpha = -2 d \ln G$ holds with the tension-RELAXING sign during the expansion phase we occupy, while the sign would have been opposite near the bounce. The programme’s sharpest present prediction therefore survives for our epoch as a special case of the cyclic history. One honest tension with data must be flagged: the observed accelerating expansion does not obviously favour a future recollapse, so a cyclic model must argue that the present acceleration is a transient phase of tension relaxation that eventually reverses — itself a falsifiable commitment.

Status. The cyclic picture is a CANDIDATE initial-state framework, not a derived history. Its internal appeal (tension regeneration; high-tension start as collapse output; a finite-density bounce) is real, but its viability as an INFINITELY cyclic model depends entirely on the entropy-reset question, and its consistency with observation depends on the present acceleration being transient. All three are labelled open.

11. Big-Bang Nucleosynthesis: Required Conditions

Big-bang nucleosynthesis is one of cosmology's firmest quantitative successes, and any framework must eventually reproduce the light-element abundances. We do not attempt an abundance calculation here. Instead we state the rope-medium conditions that would be PREREQUISITE for nuclear formation as the programme's nuclear-sector physics understands it: a phase of the network dense and energetic enough to support the nuclear-binding configurations described in the nuclear paper, followed by cooling (in the Θ sense of the thermodynamics paper) through the relevant scales.

This is deliberately modest: BBN is a place where the mesh framework must eventually MATCH a precise known result, and matching it quantitatively is future work. Naming the prerequisite conditions is the appropriate first step, and flags BBN as a strong potential FALSIFIER — if the rope medium cannot support the required sequence of conditions, that is evidence against the framework (see §16).

12. Structure Formation

Cosmic structure — the filaments, walls, and voids of the galaxy distribution — must in the rope picture arise from fluctuations in the network's own fields: tension, orientation, and defect density. The framework question is whether such fluctuations can seed and grow structure with the observed statistics.

Two features make this more than hand-waving. First, the programme's galactic-dynamics sector (MOND / radial-acceleration relation from rope-winding coherence) already ties rope properties to the dynamics of bound structures, so structure GROWTH has an existing rope handle. Second, if the initial network is critical (§10), critical fluctuations are naturally scale-invariant, which is the observed character of the primordial spectrum. Whether the growth rate and the detailed statistics match observation is unaddressed here and is explicitly future work; the rope model's MOND-like modification of gravity could either help or hurt the fit, and honestly we do not yet know which.

13. Connections to the Rest of the Programme

The cosmology sector is tightly coupled to the rest of the programme through shared rope primitives, which is both a constraint and a source of predictions:

- Thermodynamics: cosmic entropy is the configuration-degeneracy entropy of the network; the phase structure (Coulomb vs confined) governs whether electromagnetism is long-range at a given epoch.
- Electromagnetism & gravity: both couplings depend on the tension T , so the cosmic evolution of T (§4) links them — the basis of the prediction in §14.
- MOND / galactic dynamics: rope-winding coherence connects the medium to bound-structure dynamics and hence to structure formation.
- Black holes: tension-EXHAUSTION configurations (the horizon is the $T \rightarrow 0$ surface of the derived medium dictionary; corrected at GRV-034); their cosmic population and entropy connect to §6 (and to the still-open Bekenstein–Hawking derivation). Status after the v2.2.8 whisper arc: the entropy FORM is derived ($S \sim A/a^2$ on the one-way reconnection count over a derived Rindler shell, meeting the Planck-length identification of GRV-007 -- GRV-037/038); the temperature emerges from accretion-powered reconnection noise at 1.3 times $\kappa/2\pi$ (GRV-043, offset unexplained); the $1/4$ coefficient remains open, and steady evaporation of isolated holes is a registered divergence (GRV-039).
- Particle sector: the defect and knot content of the early network sets the matter content, tying nucleosynthesis and structure to particle topology.

13A. Why Cosmology Is Harder Than Local Physics

It is worth stating explicitly why the cosmological sector is, and should be, less complete than the local sectors of the programme. The local sectors — electromagnetism, gravity, particle masses — concern EQUILIBRIUM: they describe the stable configurations of the rope medium and the small fluctuations about them, for which free-energy minimisation and the effective-field-theory machinery of the thermodynamics paper apply. Cosmology is different in kind. It concerns the EVOLUTION OF EQUILIBRIUM ITSELF — how the background state $R(t)$ drifts over cosmic time, carrying the local equilibria with it.

This is why the local sectors can be derived while the cosmological one can, at this stage, only be framed. Deriving the evolution law F in $dR/dt = F(R)$ is a genuinely harder problem than characterising equilibrium, because it is a statement about non-equilibrium dynamics — the very regime the thermodynamics paper flagged as its principal open problem (the equilibrium assumption, and the undetermined nature of the fluctuation parameter Θ). The cosmological sector inherits that gap directly: until non-equilibrium rope thermodynamics is developed, F can be constrained (by conservation laws and candidate mechanisms) but not derived. Naming this cleanly is itself useful — it locates cosmology’s difficulty in a specific, identified gap rather than in vague incompleteness.

13B. A Cosmic Phase Diagram

The thermodynamics paper gave the rope medium a phase structure (ordered/Coulomb vs disordered/confined, separated by defect proliferation). Cosmology can be pictured as a

TRAJECTORY through that phase structure as the state vector $R(t)$ evolves. Figure 1 sketches a possible such trajectory. We stress what it is and is not: it is a hypothesised thermodynamic evolution consistent with the framework, NOT a claimed history. Each transition is a conjecture to be derived from the (still-undetermined) evolution law F , or falsified.

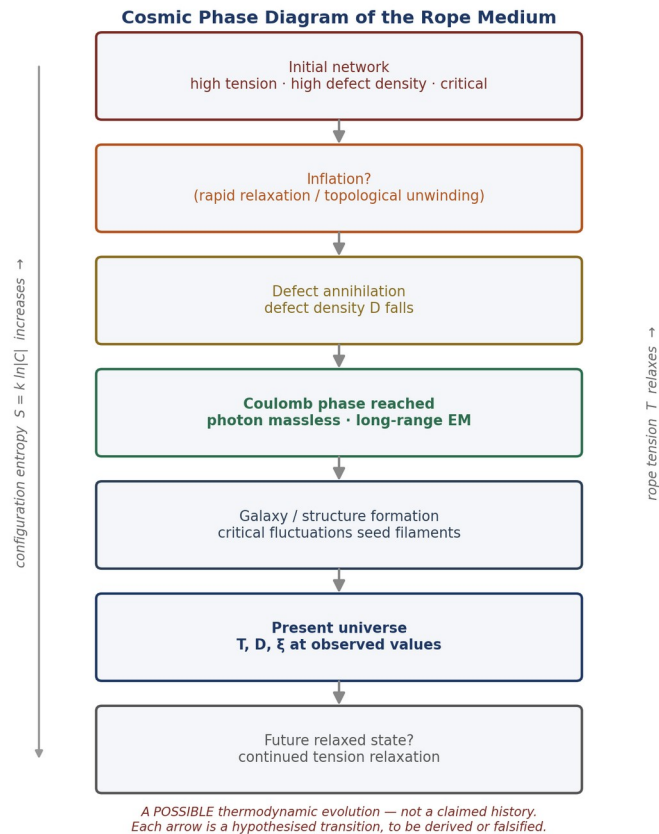


Figure 1. A possible thermodynamic trajectory of the rope medium: from a high-tension, high-defect critical initial network, through defect annihilation into the Coulomb (massless-photon) phase where long-range electromagnetism and structure formation become possible, to the present and a possible future relaxed state. Entropy increases and tension relaxes along the trajectory. This is possible evolution, not claimed history — each arrow is a hypothesised transition to be derived or falsified.

The value of the diagram is that it connects the cosmological sector directly to the thermodynamics paper: cosmic history becomes a path through the SAME phase diagram that governs local electromagnetism, and the epoch at which the Coulomb phase is reached is the epoch at which long-range electromagnetism (and hence structure and observers) becomes possible. It also makes the framework's logic vivid: if the medium is relaxing along this trajectory, the constants that depend on T are drifting with it (§14).

14. Predictions

We separate the one structurally-forced prediction from the conditional and the speculative, following the discipline of the falsifiable-predictions paper.

14.1 *Cosmic expansion drives correlated drift of the constants (the centrepiece)*

This is the cosmology sector's most valuable contribution, because it converts an abstract relation into a cosmological one. The programme already predicts that α and G are not independent, since both depend on the rope tension T ($\alpha \sim 2T^2/(\kappa a)$; gravity governed by T). IF cosmic expansion corresponds even partly to relaxation of the rope tension (§4's tension candidate), THEN the fundamental constants must drift over cosmic time in the correlated, scale-free way the programme predicts:

$$d \ln \alpha = -2 d \ln G \quad (\text{tension-driven}) \quad - \quad \text{now as a function of cosmic epoch} \quad (14.1)$$

Stated at the correct strength. The STRUCTURAL prediction — that cosmic expansion, IF tension-driven, forces correlated time-variation of α and G rather than independent drift — is the robust content, and it is genuinely risky: it is testable against the same quasar- α and pulsar/lunar-laser- $\dot{G}$ bounds, now interpreted as cosmic-time drifts. The specific coefficient -2 remains provisional pending a rigorous $G(T, \kappa, a)$, and the LINK to expansion is itself conditional on the tension-relaxation mechanism being the correct reading of expansion. We foreground the structural claim and flag both conditionals plainly.

This cuts both ways, which is what makes it science: current bounds on constant-variation are tight. If those bounds are eventually shown to EXCLUDE the correlated drift that tension-driven expansion would require, that would be evidence AGAINST the tension-relaxation reading of expansion — a genuine falsification route, not just a confirmation opportunity (§16).

The hinge, stated once more

This entire prediction hangs on ONE conditional: that cosmic expansion corresponds PRIMARILY to relaxation of the rope tension T . If expansion is instead mostly increasing rope spacing, defect annihilation, or network reorganisation (the other candidates of §4), the α - G link weakens or changes form. The correlated-drift prediction is therefore only as strong as the tension-relaxation reading of expansion — which is itself an open question. We state this plainly because it is the load-bearing assumption of the sector's sharpest claim.

14.2 *Further predictions*

- Cosmological birefringence: a residual large-scale rope-orientation structure would rotate CMB and distant-source polarisation — connecting to the existing birefringence prediction and to §9.
- Defect-density signatures: relic topological defects (heavy monopoles, and possibly cosmic-string-like rope structures) with abundances set by the early network.
- Modified structure growth: the rope/MOND modification of gravity implies a growth history potentially distinguishable from Λ CDM.

- Time-dependent dark-energy equation of state $w(t) \neq -1$, if dark energy is defect pressure or configuration-entropic (§7).

15. Status of Each Result

Every element of this paper is labelled by status. For a framework paper the honest distribution is weighted toward Open Problem — that is the accurate picture of a cosmological sector at inception, and stating it plainly is the point.

Element	Status
Cosmological principle for rope properties	Modeling Assumption (reformulation of the standard one)
Cosmological rope state variables	Framework Definition
Conservation-law structure (energy/topology/defects)	Framework (posited, not solved)
Configuration-entropy definition of cosmic entropy	Derived (inherits the thermodynamics definition)
Expansion mechanism (which of the four)	Open Problem
Dark energy source	Open Problem (candidates only; magnitude unexplained)
Inflation mechanism	Open Problem (candidates + acceptance test)
CMB / last-scattering account	Open Problem (signature identified: polarisation)
Initial-state identity	Open Problem (critical-network favoured, not derived)
BBN	Open Problem (prerequisite conditions stated; strong potential falsifier)
Structure formation	Open Problem (existing MOND handle; fit unknown)
α -G correlated cosmic drift	Structural prediction (conditional on tension-driven expansion; coefficient provisional)
Cosmological birefringence	Conditional prediction (on residual orientation structure)

16. Falsifiability: What Would Refute Rope Cosmology

A cosmological framework that could not be wrong would not be science. The rope-cosmology picture is exposed to refutation at several points, and we state them explicitly — this is a strength, not a liability:

1. Constant-variation bounds: if α and G are shown to drift independently, or if their variation is excluded at levels the tension-driven picture requires, the tension-relaxation reading of expansion (§14.1) is falsified.

2. Dark-energy equation of state: a precise measurement of $w = -1$ (pure cosmological constant, no time dependence) would exclude the defect-pressure and entropic candidates of §7, leaving only the less-predictive residual-tension option.
3. CMB polarisation: absence of any rope-orientation birefringence signature, at improving sensitivity, progressively constrains the residual large-scale orientation structure toward zero.
4. BBN: if the rope medium cannot support the sequence of conditions required for the observed light-element abundances, that is direct evidence against the framework.
5. Structure growth: if the rope/MOND-modified growth history is excluded by large-scale-structure surveys, the galactic and cosmological sectors are jointly constrained.

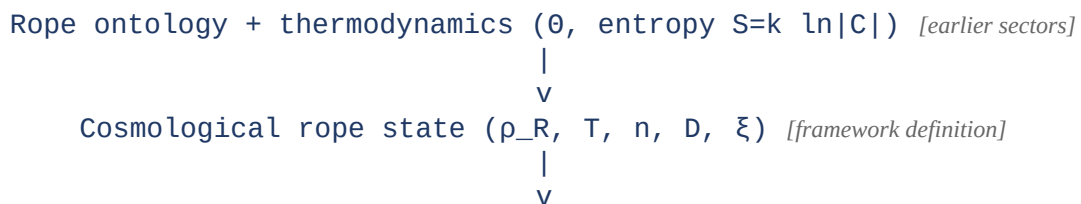
We regard the constant-variation route (1) as the sharpest near-term test, because it connects a cosmological hypothesis (tension-driven expansion) to precision laboratory and astrophysical bounds that already exist and are improving.

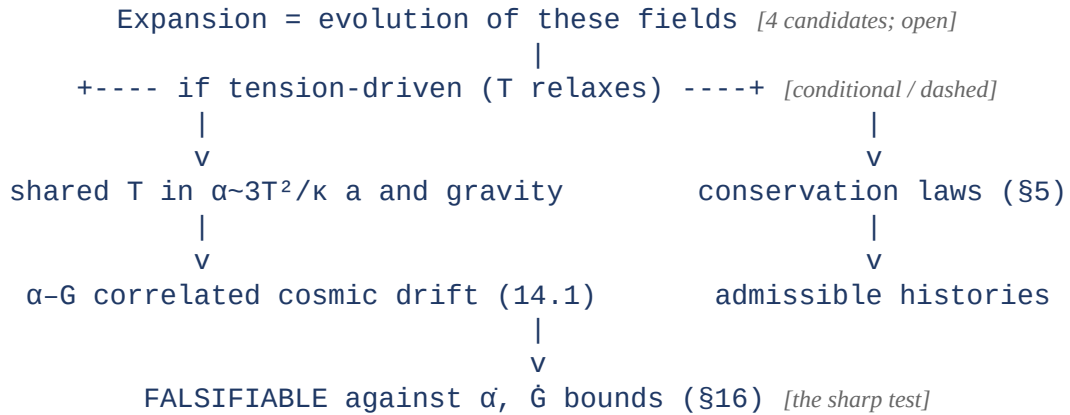
17. The Natural Successor: Dynamics of the Cosmological Rope State

This framework paper deliberately stops at defining the state vector $R = (\rho_R, T, n, D, \xi)$ and the form $dR/dt = F(R)$ without deriving F . The natural — and we believe correct — next step is not a paper on inflation, dark energy, or the CMB individually, but a single paper whose sole purpose is to derive the evolution law F for R from the conservation structure of §5 and the (to-be-developed) non-equilibrium thermodynamics of the medium. Once F exists, the classic cosmological topics cease to be separate stories and become CONSEQUENCES of one dynamical system: expansion history, the dark-energy equation of state, the inflationary epoch, and structure growth would all be read off from solutions of $dR/dt = F(R)$. We flag “Dynamics of the Cosmological Rope State” as the priority successor, and regard the present paper’s main achievement as making that successor a well-posed problem rather than an open-ended ambition.

Appendix A. Dependency Graph

How the cosmological claims depend on earlier sectors and on the choice of expansion mechanism. The dashed link is conditional — it holds only if expansion is tension-driven — and is the hinge on which the sector’s sharpest prediction turns.





The graph makes the logical structure explicit: almost everything cosmological is currently open, but one conditional branch — tension-driven expansion → correlated constant-drift → existing bounds — is a genuine, falsifiable line from the ontology to data. That single branch is the cosmology sector’s present scientific content; the rest is well-posed open problems.

Conclusion

This paper establishes a conceptual and structural framework for a cosmological sector within the Mesh Programme. It defines the cosmological state of the rope medium, enumerates candidate mechanisms for expansion, dark energy, and inflation together with their distinguishing signatures, extends the configuration-entropy account to cosmic scales, and states explicit falsification conditions. It does NOT derive the expansion history or explain dark energy, inflation, the CMB, or nucleosynthesis; for those, the honest status is well-posed open problems, labelled as such. Its one piece of genuine present scientific content is the conditional but falsifiable link between tension-driven expansion and the correlated cosmic drift of α and G — a prediction that ties the cosmological sector to precision bounds that already exist. Quantitative comparison with cosmological observations is reserved for future work; the contribution here is to pose the cosmological questions in rope-medium terms precisely enough that such comparison, and such refutation, becomes possible.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.